STAT 6206: Bayesian Statistics with Applications in Machine Learning

Assignment 2

Academic year 24/25, second term

Due date: April 9, 2025

1. The Bovine Toll-like receptor data (BTdata) is a dataset collected from a study involving cows and their offspring. It contains multiple observations on the following variables:

   - **dam**(j): A factor identifying the dam.
   - **tarsus**(Y_{ij}): A numeric vector describing the tarsus length of the offspring at a given time point.
   - **sex**(i): A factor indicating the sex of the offspring.

   The linear mixed model is defined as:

   $$Y_{ij} = \beta_0 + \beta_1 \text{sex}_{ij} + u_{dam(j)} + \epsilon_{ij}$$

   where $u_{dam(j)}$ is the random effect from the dam, assumed to be $u_{dam(j)} \sim N(0, \sigma^2)$.

   (a) Load the data by using the 'BTdata.txt' dataset. Use R2Winbugs or another programming language to fit the linear mixed model. Report the parameter estimate.

   (b) Interpret the parameter estimates.

2. Let

   - y_{it} — time to falling asleep for subject i at occasion t,
   - t — occasion (0 = initial, 1 = follow-up),
   - x — treatment (0 = placebo, 1 = active treatment).

The cumulative logit model with random intercept:

$$\text{logit}[p(y_{it} \leq j|u_i)] = u_i + \alpha_j + \beta_1 t + \beta_2 x + \beta_3 tx,$$

where u_i is a random effect, and $u_i \sim N(0, \sigma^2)$.

(a) Load the data in Insomnia.txt. Use R2Winbugs or another programming language to fit this model. Report the parameter estimate.

(b) Interpret the parameter estimates.